

exercises

July 11, 2026

1 Infinite Series, Power Series

```
[32]: %load_ext autoreload
      %autoreload 2

      import pandas as pd
      import numpy as np
      import matplotlib.pyplot as plt
      import math
      from IPython import display
      from IPython.display import Markdown
      from scipy.special import factorial
      from scipy import stats
      from scipy import constants

      # Below are some adjustments:
      # from IPython.display import HTML
      # HTML('<style>img[alt="skan"] { filter: invert(100%); }</style>')
      # plt.style.use('dark_background')
```

The autoreload extension is already loaded. To reload it, use:

```
%reload_ext autoreload
```

1.1 Problems, Section 2

In the following problems, find the limit of the given sequence as $n \rightarrow \infty$.

1.1.1 2.6

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!}$$

```
[33]: # 1. Simulation setup
      # We keep the range small (1 to 10) because  $n^n$  grows insanely fast.
      n_max: float = 10.0
      step: float = 0.1
      start: float = 1.0
      n_values = np.arange(start, n_max, step)
```

```

custom_mk = 10

numerators = n_values**n_values
denominators = factorial(n_values)
results = numerators / denominators

# 3. Plotting the chart
plt.figure(figsize=(10, 6))

# Draw the main line
plt.plot(n_values, results, color='red', marker='o', markevery=custom_mk,
        ↪label=r'values $\frac{n^n}{n!}$')

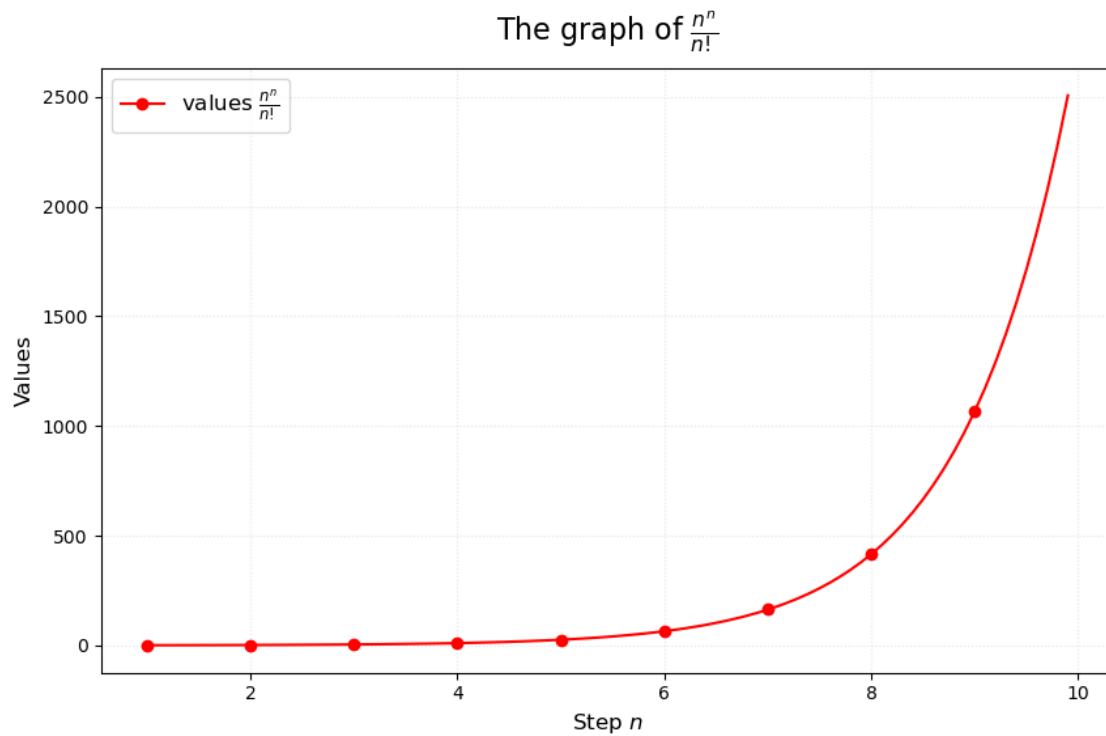
# You may ask "why is this code excluded?"
# Let's have a look at the table of Values below
# It's reasonable not to include numerator and denominator because of the rapid
    ↪growth of those values
#plt.plot(n_values, numerators, color='blue', linestyle='-', marker='s',
    ↪markevery=5, label=r'numerator $n^n$')
#plt.plot(n_values, denominators, color='green', linestyle='--', marker='^',
    ↪markevery=5, label=r'denominator $n!$')

# Aesthetics and labels
plt.title(r'The graph of $\frac{n^n}{n!}$', fontsize=16, pad=15)
plt.xlabel('Step $n$', fontsize=12)
plt.ylabel('Values', fontsize=12)
plt.grid(True, linestyle=':', alpha=0.3)
plt.legend(fontsize=12)

# Render the plot
plt.show()

table_data = {
    r"Step $n$": n_values,
    r"Numerator $n^n$": numerators,
    r"denominator $n!$": denominators,
    r"Value $\frac{n^n}{n!}$": results
}
df = pd.DataFrame(table_data)
df_sliced = df.iloc[::10]
Markdown(df_sliced.to_markdown(index=False))

```



[33] :

Step n	Numerator n^n	denominator $n!$	Value $\frac{n^n}{n!}$
1	1	1	1
2	4	2	2
3	27	6	4.5
4	256	24	10.6667
5	3125	120	26.0417
6	46656	720	64.8
7	823543	5040	163.401
8	1.67772e+07	40320	416.102
9	3.8742e+08	362880	1067.63

$$\lim_{n \rightarrow \infty} \frac{n^n}{n!} = \infty$$

1.1.2 2.7

$$\lim_{n \rightarrow \infty} \left((1 + n^2)^{\frac{1}{\ln n}} \right)$$

Simulation:

```
[34]: def compute():
    n_max = 20
    n_start = 2.0
    n_step = 0.1
    n_values = np.arange(n_start, n_max, n_step)
    results = (1 + n_values**2)**(1 / np.log(n_values))
    powers = 1 / np.log(n_values)
    bases = 1 + n_values**n_values

    plt.figure(figsize=(10,6))

    plt.plot(n_values, results, color='red', linestyle="-", marker='o',
    ↪markevery=10, label=r'Values  $(1+n^2)^{\frac{1}{\ln n}}$ ')

    euler_value = np.exp(2)

    plt.axhline(y=euler_value, color='blue', linestyle='--',linewidth=2,
    ↪label=r'$e^2$')

    plt.title(r"The graph of  $(1+n^2)^{\frac{1}{\ln n}}$ ", fontsize=16, pad=15)
    plt.xlabel('$n$', fontsize=14)
    plt.ylabel('Values', fontsize=14)
    plt.grid(True, linestyle=':', alpha=0.5)
    plt.legend(fontsize=14, loc='upper right')

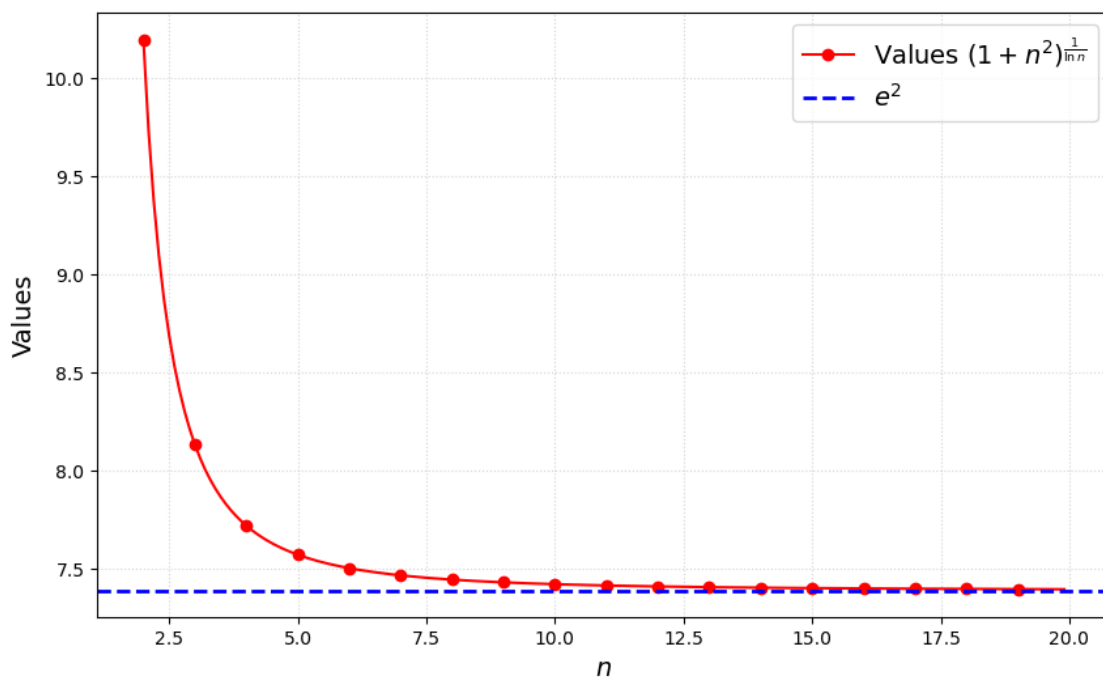
    plt.show()

    table_data = {
        r"Step $n$": n_values,
        r"Base  $1 + n^2$ ": bases,
        r"Power  $\frac{1}{\ln n}$ ": powers,
        r"Values  $(1+n^2)^{\frac{1}{\ln n}}$ ": np.round(results, 4)
    }

    df = pd.DataFrame(table_data)
    df_sliced = df.iloc[:20]
    return Markdown(df_sliced.to_markdown(index=False))

compute()
```

The graph of $(1 + n^2)^{\frac{1}{\ln n}}$



[34] :

Step n	Base $1 + n^2$	Power $\frac{1}{\ln n}$	Values $(1 + n^2)^{\frac{1}{\ln n}}$
2	5	1.4427	10.1953
4	257	0.721348	7.7194
6	46657	0.558111	7.5029
8	1.67772e+07	0.480898	7.4444
10	1e+10	0.434294	7.4211
12	8.9161e+12	0.40243	7.4097
14	1.1112e+16	0.378923	7.4033
16	1.84467e+19	0.360674	7.3995
18	3.93464e+22	0.345976	7.3969

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} (1 + n^2)^{\frac{1}{\ln n}} = \left[\infty^0 \right] \\
 & (1 + n^2)^{\frac{1}{\ln n}} = e^{\ln \left[(1 + n^2)^{\frac{1}{\ln n}} \right]} = e^{\frac{\ln (1 + n^2)}{\ln n}} = e^{\frac{\ln n^2 + \ln \left[1 + \frac{1}{n^2} \right]}{\ln n}} \\
 & \lim_{n \rightarrow \infty} (1 + n^2)^{\frac{1}{\ln n}} = \lim_{n \rightarrow \infty} e^{\frac{\ln n^2 + \ln \left[1 + \frac{1}{n^2} \right]}{\ln n}} = \lim_{n \rightarrow \infty} e^{\frac{2 \ln n + \ln 1}{\ln n}} = e^2
 \end{aligned}$$

1.1.3 2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

```
[35]: def compute():
    n_max = 10
    n_step = 0.1
    n_start = 2
    n_values = np.arange(n_start, n_max, n_step)

    numerators = factorial(n_values)**2
    denominators = factorial(2*n_values)
    results = numerators / denominators

    plt.figure(figsize=(16,9))

    plt.plot(n_values, results, 'o-r', markevery=10, linewidth=3,
    ↪label=r'Values $\frac{(n!)^2}{(2n)!}$')
    #plt.plot(n_values, numerators, color='green', marker='^', linestyle='-',
    ↪markevery=11, label=r'Numerators $(n!)^2$')

    plt.title(r'the graph of: $\frac{(n!)^2}{(2n)!}$', fontsize=30, pad=15)
    plt.xlabel(r"$n$", fontsize=16)
    plt.ylabel(r"Values", fontsize=16)
    plt.grid(True, linestyle=':', alpha=0.5)
    plt.legend(fontsize=16, loc='upper right')

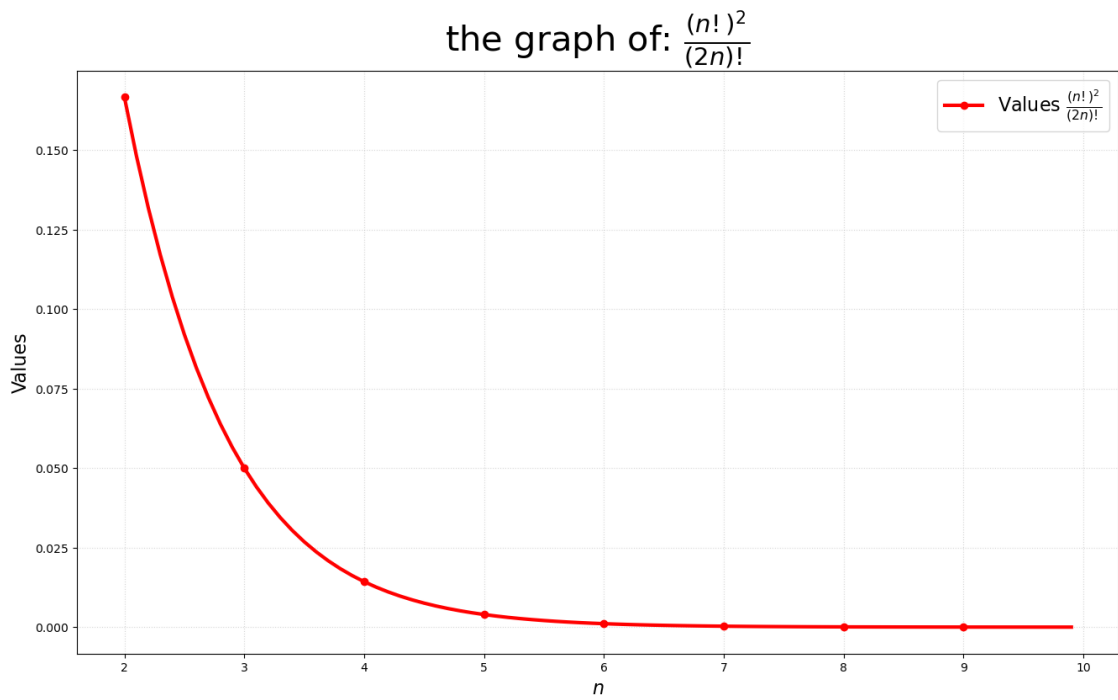
    plt.show()

    table_data = {
        r'Step $n$': n_values,
        r'Numerator $(n!)^2$': numerators,
        r'Denominator' $(2n)!$: denominators,
        r"Values $\frac{(n!)^2}{(2n)!}$": results
    }

    df = pd.DataFrame(table_data)
    df_sliced = df.iloc[:10]

    return Markdown(df_sliced.to_markdown(index=False))

compute()
```



[35] :

Step n	Numerator $(n!)^2$	Denominator' $(2n)!$	Values $\frac{(n!)^2}{(2n)!}$
2	4	24	0.166667
3	36	720	0.05
4	576	40320	0.0142857
5	14400	3.6288e+06	0.00396825
6	518400	4.79002e+08	0.00108225
7	2.54016e+07	8.71783e+10	0.000291375
8	1.6257e+09	2.09228e+13	7.77001e-05
9	1.31682e+11	6.40237e+15	2.05677e-05

2.8

$$\lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!}$$

$$(n!)^2 = (1 \cdot 2 \cdot 3 \cdot \dots \cdot n)(1 \cdot 2 \cdot 3 \cdot \dots \cdot n)$$

$$(2n)! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n(n+1) \cdot (n+2) \cdot \dots \cdot 2n$$

$$a_n = \frac{n! \cdot n!}{n! \cdot (n+1)(n+2) \dots 2n} = \frac{n!}{(n+1)(n+2) \dots 2n} =$$

$$= \frac{1}{n+1} \cdot \frac{2}{n+2} \cdot \frac{3}{n+3} \dots \frac{n}{n+n}$$

$$a_n \leq \left(\frac{1}{2}\right)^n \Rightarrow \lim_{n \rightarrow \infty} \frac{(n!)^2}{(2n)!} = 0$$

1.1.4 2.9

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right)$$

```
[36]: def simple_plot(n_start: float, n_max: float, n_step: float, title: ):
      n_values = np.arange(n_start, n_max, n_step)
      results = n_values * np.sin(1/n_values)
```

Cell In[36], line 1

```
def simple_plot(n_start: float, n_max: float, n_step: float, title: ):
    ^
```

SyntaxError: invalid syntax

2.9

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = [0 \cdot \infty]$$

$\nearrow \infty$
 $\searrow 0$

$x = \frac{1}{n} \Rightarrow n = \frac{1}{x}$
 As $n \rightarrow \infty$, $\frac{1}{x}$ approaches 0

$$\lim_{n \rightarrow \infty} n \sin \frac{1}{n} = \lim_{x \rightarrow 0} \frac{1}{x} \sin x = \lim_{x \rightarrow 0} \frac{\sin x}{x} =$$

$$= \lim_{n \rightarrow \infty} 1 \cdot \frac{\sin x}{x} = 1$$

1.2 Problems, Section 4

For the following series, write formulas for the sequences a_n , S_n , and R_n , and find the limits of the sequences as $n \rightarrow \infty$ (if the limits exist).

1.2.1 4.1

$$\sum_1^{\infty} \frac{1}{2^n}$$

h.1.

$$\sum_{n=1}^{\infty} \frac{1}{2^n}$$

$$a_n = \frac{1}{2^n}$$

$$S_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^n}$$

$$a_1 = \frac{1}{2} \quad \left. \begin{array}{l} r = \frac{1}{2} \end{array} \right\} S_n = \frac{a_1 \cdot (1 - r^n)}{1 - r} = \frac{\frac{1}{2} [1 - (\frac{1}{2})^n]}{1 - \frac{1}{2}}$$

$$S_n = \frac{\frac{1}{2} - (\frac{1}{2})^{n+1}}{\frac{1}{2}} = 1 - (\frac{1}{2})^n$$

$$R_n = S - S_n = \lim_{n \rightarrow \infty} S_n - S_n =$$

$$= 1 - 1 + (\frac{1}{2})^n = (\frac{1}{2})^n$$

1.2.2 4.2

$$\sum_{n=0}^{\infty} \frac{1}{5^n}$$

4.2

$$\sum_0^{\infty} \frac{1}{5^n} = 1 + \frac{1}{5} + \frac{1}{25} + \dots$$

I. $a_n = \frac{1}{5^{n-1}}$ *usually, geometric progression starts with $n=1$, but in this case $a_1=1$, so we need to subtract 1 from n*

II. $a_1=1$ $r = \frac{1}{5}$ $\Rightarrow S_n = 1 \cdot \frac{[1 - (\frac{1}{5})^n]}{1 - \frac{1}{5}} = \frac{1 - (\frac{1}{5})^n}{\frac{4}{5}} = \frac{5 - (\frac{1}{5})^{n-1}}{4} = \frac{5}{4} [1 - (\frac{1}{5})^n]$

III. $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[\frac{5}{4} [1 - (\frac{1}{5})^n] \right] = \left[\text{as } n \rightarrow \infty, (\frac{1}{5})^n \rightarrow 0 \right] = \frac{5}{4}$

IV. $R_n = S - S_n = \frac{5}{4} - \frac{5}{4} [1 - (\frac{1}{5})^n] = \frac{5}{4} - \frac{5}{4} + \frac{5}{4} (\frac{1}{5})^n = \frac{1}{4} \cdot (\frac{1}{5})^{n-1}$

V. $\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{4} \cdot (\frac{1}{5})^{n-1} = \left[\text{as } n \rightarrow \infty, (\frac{1}{5})^{n-1} \rightarrow 0 \right] = 0$

1.2.3 4.3

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} \dots$$

4.3

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16}$$

I. $a_n = \frac{1}{(-2)^{n-1}}$

II. $a_1=1$ $r = -\frac{1}{2}$ $\Rightarrow S_n = 1 \cdot \frac{1 - (-\frac{1}{2})^n}{1 - (-\frac{1}{2})} = \frac{1 - (-\frac{1}{2})^n}{\frac{3}{2}} = \frac{2}{3} [1 - (-\frac{1}{2})^n]$

III. $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{2}{3} [1 - (-\frac{1}{2})^n] = \left[\text{as } n \rightarrow \infty, (-\frac{1}{2})^n \rightarrow 0 \right] = \frac{2}{3}$

IV. $R_n = S - S_n = \frac{2}{3} - \frac{2}{3} [1 - (-\frac{1}{2})^n] = \frac{2}{3} (-\frac{1}{2})^n$

V. $\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{2}{3} (-\frac{1}{2})^n = \left[\text{as } n \rightarrow \infty, (-\frac{1}{2})^n \rightarrow 0 \right] = 0$

1.2.4 4.4

$$\sum_1^{\infty} e^{-n \ln 3}$$

$$\begin{aligned}
 & \sum_{n=1}^{\infty} e^{-n \ln 3} \quad \nearrow \ln 3 = \log_e 3, \quad x = e^{\ln x} \\
 & e^{-n \ln 3} = (e^{-\ln 3})^n = \left(\frac{1}{3}\right)^n \\
 \text{I} \quad & a_n = \left(\frac{1}{3}\right)^n \\
 \text{II} \quad & a_1 = \frac{1}{3} \quad \left. \begin{array}{l} r = \frac{1}{3} \end{array} \right\} S_n = \frac{1}{3} \cdot \frac{1 - \left(\frac{1}{3}\right)^n}{1 - \frac{1}{3}} = \frac{1}{2} \cdot \cancel{\frac{2}{2}} \cdot \left[1 - \left(\frac{1}{3}\right)^n\right] = \frac{1}{2} \left[1 - \left(\frac{1}{3}\right)^n\right] \\
 \text{III} \quad & \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{1}{2} \left[1 - \left(\frac{1}{3}\right)^n\right] = \left[\text{as } n \rightarrow \infty, \left(\frac{1}{3}\right)^n \rightarrow 0 \right] = \frac{1}{2} \\
 \text{IV} \quad & R_n = S - S_n = \frac{1}{2} - \frac{1}{2} \left[1 - \left(\frac{1}{3}\right)^n\right] = \frac{1}{2} \left(\frac{1}{3}\right)^n \\
 \text{V} \quad & \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{1}{3}\right)^n = \left[\text{as } n \rightarrow \infty, \left(\frac{1}{3}\right)^n \rightarrow 0 \right] = 0
 \end{aligned}$$

1.2.5 4.5

$$\sum_0^{\infty} e^{2n \ln \sin\left(\frac{\pi}{3}\right)}$$

$$4.5 \quad \sum_0^{\infty} e^{2n \ln \sin \frac{\pi}{3}}$$

$$e^{2n \ln \sin \frac{\pi}{3}} = e^{2n \ln \frac{\sqrt{3}}{2}} = \left[e^{2 \ln \frac{\sqrt{3}}{2}} \right]^n = \left[e^{\ln \frac{3}{4}} \right]^n = \left(\frac{3}{4} \right)^n$$

I $a_n = \left(\frac{3}{4} \right)^{n-1}$

II $a_1 = 1$
 $r = \frac{3}{4} \quad \left. \vphantom{\begin{matrix} a_1 = 1 \\ r = \frac{3}{4} \end{matrix}} \right\} S_n = 1 \cdot \frac{1 - \left(\frac{3}{4} \right)^n}{1 - \frac{3}{4}} = \frac{1 - \left(\frac{3}{4} \right)^n}{\frac{1}{4}} = 4 \left[1 - \left(\frac{3}{4} \right)^n \right]$

III $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} 4 \left[1 - \left(\frac{3}{4} \right)^n \right] = \left[\text{as } n \rightarrow \infty, \left(\frac{3}{4} \right)^n \rightarrow 0 \right] = 4$

IV $R_n = S - S_n = 4 - 4 \left[1 - \left(\frac{3}{4} \right)^n \right] = 4 \left(\frac{3}{4} \right)^n$

V $\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} 4 \left(\frac{3}{4} \right)^n = \left[\text{as } n \rightarrow \infty, \left(\frac{3}{4} \right)^n \rightarrow 0 \right] = 0$

1.2.6 4.6

$$\sum_1^{\infty} \frac{1}{n(n+1)}$$

h. 6

$$\sum_1^{\infty} \frac{1}{n(n+1)} = \sum_1^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right]$$

$$\left. \begin{array}{l} a_1 = 1 - \frac{1}{2} = \frac{1}{2} \\ a_2 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \\ a_3 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \\ a_n = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \\ a_5 = \frac{1}{5} - \frac{1}{6} = \frac{1}{30} \end{array} \right\} \begin{array}{l} I \ a_n = \frac{1}{n(n+1)} \\ \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n(n+1)} = 0 \end{array}$$

$$II \quad \sum_1^{\infty} \left[\frac{1}{n} - \frac{1}{n+1} \right] = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{n+1} \quad \text{as } n \rightarrow \infty$$

$$S_n = 1 - \frac{1}{n+1}$$

$$III \quad \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[1 - \frac{1}{n+1} \right] = \left[\text{as } n \rightarrow \infty, \frac{1}{n+1} \rightarrow 0 \right] = 1$$

$$IV \quad R_n = S - S_n = 1 - 1 + \frac{1}{n+1} = \frac{1}{n+1}$$

$$V \quad \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$$

1.2.7 4.7

$$\frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \frac{7}{3 \cdot 4} - \frac{9}{4 \cdot 5} + \dots$$

h. 7

$$\frac{3}{1 \cdot 2} - \frac{5}{2 \cdot 3} + \frac{7}{3 \cdot 4} - \frac{9}{4 \cdot 5} + \dots =$$

$$\text{I } a_n = \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)}, \quad \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)} = 0$$

$$\text{II } S_n = S_b - S_c$$

$$S_b = \frac{3}{1 \cdot 2} + \frac{7}{3 \cdot 4} + \frac{11}{5 \cdot 6} + \dots = \frac{3}{1} - \frac{3}{2} + \frac{7}{3} - \frac{7}{4} + \frac{11}{5} - \frac{11}{6} + \dots$$

$$S_c = \frac{5}{2} - \frac{5}{3} + \frac{9}{4} - \frac{9}{5} + \frac{13}{6} - \frac{13}{7} + \dots$$

$$S_n = 3 - \frac{3}{2} - \frac{5}{2} + \frac{7}{3} + \frac{5}{3} - \frac{7}{4} - \frac{9}{4} + \frac{11}{5} + \frac{9}{5} - \frac{11}{6} - \frac{13}{6} + \dots$$

$$S_n = 3 - \frac{8}{2} + \frac{12}{3} - \frac{16}{4} + \frac{20}{5} - \frac{24}{6} + \dots$$

$$S_n = 3 - 4 + 4 - 4 + 4 - 4 + 4 - 4 + \dots + \frac{(-1)^{n-1} \cdot (2n+1)}{n(n+1)}$$

$$S_n = 3 + (-1)^{n-1} \cdot \frac{(2n+1)}{n(n+1)}$$

$$\text{III } \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left[3 + (-1)^n \frac{2n+1}{n(n+1)} \right] = \left[\text{as } n \rightarrow \infty, \frac{2n+1}{n(n+1)} \rightarrow 0 \right] = 3$$

$$\text{IV } R_n = S - S_n = 3 - 3 + \frac{(-1)^n (2n+1)}{n(n+1)} = \frac{(-1)^n (2n+1)}{n(n+1)}$$

$$\text{V } \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} \frac{(-1)^n (2n+1)}{n(n+1)} = \left[\text{as } n \rightarrow \infty, \frac{2n+1}{n(n+1)} \rightarrow 0 \right] = 0$$